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Spatially heterogeneous organic carbon remineralization as a driver for highly negative carbonate δ^{13}C excursions

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ABSTRACT

The carbon isotopic composition of marine carbonate records ancient carbon cycling throughout Earth history. However, local depositional processes and diagenesis introduce variability that can obscure or mimic global signals. Such effects complicate interpretations of ancient successions, where carbonate δ^{13}C may reflect local processes rather than global changes in the oceanic carbon reservoir. Here, we examine the Ediacaran Shuram carbon isotope excursion, a globally observed, highly negative and spatially variable δ^{13}C anomaly associated with the diversification of multicellular life. We test whether the incorporation of carbon derived from organic matter degradation into bulk carbonate sediment can produce the stratigraphic distribution and magnitude of δ^{13}C changes observed globally. We simulate the transport and early diagenesis of carbonates in a basin experiencing an oscillating sea level cycle. Total organic carbon (TOC) in sediments is greatest at the slope and shelf margin where productivity and accumulation rates are highest. We demonstrate that local incorporation of carbon with low δ^{13}C from organic carbon respiration into bulk sediment can reproduce the magnitudes and sequence stratigraphic patterns observed in Shuram-bearing strata. Moreover, the imposed shallow-to-deep gradient in TOC accumulation generates facies-specific trends like those in the rock record, with deep water sediments consistently more negative than shallower equivalents. This mechanism offers a possible explanation for negative carbon isotope excursions without requiring coincident global changes in DIC. However, the largest excursions, like the Shuram, would require TOC accumulation at the sediment-water interface comparable to highly productive upwelling zones, roughly five times greater than typical passive margin slope settings.

1. Introduction

Shallow water carbonate sediments are archives of surface climate and marine biogeochemistry on Earth over geologic timescales (e.g., Veizer et al., 1999). In particular, the carbon isotopic composition of marine carbonates ($\delta^{13}C_{carb}$) can indicate changes in the global carbon cycle and serve as a chronostratigraphic tool to correlate basins globally (e.g., Halverson et al., 2005). Such inferences are made under the assumption that the geochemical composition of carbonate sediments faithfully preserves the δ^{13}C composition of a well mixed dissolved inorganic carbon (DIC) pool of the global ocean (e.g., Kump and Arthur, 1999). Changes in measured $\delta^{13}C_{carb}$ values are thus thought to be driven by shifts in the input and output fluxes of carbon to Earth's surface. Such shifts can be connected to major changes to Earth's ocean-atmosphere system (Kump and Arthur, 1999; Zachos et al., 2001).

Though the δ^{13}C values of open-marine carbonates may serve as accurate records for marine DIC δ^{13}C values (Zachos et al., 2001), the δ^{13}C composition of shallow platform carbonates can be significantly altered by local processes such as groundwater discharge (Patterson and Walter, 1994) or enhanced productivity (Geyman and Maloof, 2019). These local processes cause δ^{13}C values of DIC in shallow platform waters to deviate from global DIC (e.g., Swart, 2008). Beyond such local changes in the DIC pool, from which shallow water carbonates form, bulk-rock δ^{13}C can also diverge from its original value due to early diagenetic reactions. Processes such as recrystallization (e.g., aragonite and high-magnesium calcite converting to low-magnesium calcite), dolomitization, and interactions with meteoric or marine fluids can alter the isotopic composition of carbonate sediments in ways that are often difficult to detect in natural samples (Allan and Matthews, 1982; Higgins et al., 2018; Ahm et al., 2018). Additionally, the early precipitation of authigenic cements within unfilled pore spaces can incorporate isotopically light carbon from organic matter respiration, producing highly negative δ^{13}C values (Schrag et al., 2013). When these diagenetic effects are coupled with sea level change and sediment transport, diagenetically

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altered sediments can migrate across shallow platforms, redistributing isotopic variability in ways that further complicate stratigraphic records (van Wieren et al., 2026).

It is possible that these diagenetic processes can play a role in generating the highly negative δ^{13}C values observed in some ancient carbonates. While marine carbonates through much of Earth history broadly record values near 0‰ (Kump and Arthur, 1999; Zachos et al., 2001), intervals with values lower than –10 to –12‰ have been documented (Yang et al., 2021). Such excursions are difficult to reconcile with canonical models of the global carbon cycle, in which carbon inputs to the ocean-atmosphere system carry δ^{13}C values near –5‰, which would be the theoretical minimum value for marine carbonates (Kump and Arthur, 1999). Such values require additional inputs of carbon with low $\delta^{13}C_{carb}$ values, such as from organic matter respiration, processes that also have implications for organic carbon burial and Earth's redox evolution. A clear example of an enigmatic excursion in carbonate δ^{13}C values is found in sediments of the Ediacaran Period (635-538.8 Ma), known as the Shuram excursion. The Shuram is the largest δ^{13}C excursion in Earth history, with values lower than –12‰, expressed across multiple paleocontinents (e.g., Burns and Matter, 1993; Calver, 2000; Jiang et al., 2007; Bergmann et al., 2011; Macdonald et al., 2013). It also displays substantial spatial variability and consistent sequence stratigraphic patterns, with sharp downturns in δ^{13}C values during transgression followed by gradual recovery through regression (e.g., Fike et al., 2006; Husson et al., 2015b; Busch et al., 2022; Bergmann et al., 2025).

The origin of the Shuram excursion remains debated. Global changes in DIC have been argued for due to its documentation across multiple paleocontinents, and apparent synchroneity (e.g., Husson et al., 2015b; Gong and Li, 2020; Rooney et al., 2020; Busch et al., 2022). Proposed drivers for global DIC change include oxidation of methane clathrates or a large dissolved organic carbon reservoir (Bjerrum and Canfield, 2011; Rothman et al., 2003). Others attribute the excursion to diagenetic overprints, such as interaction with meteoric fluids (Knauth and Kennedy, 2009; Zhao et al., 2020), burial brines (Derry, 2010), or authigenic carbonate formation, in which pore-space precipitation of new carbonate phases incorporates DIC with low δ^{13}C values (Schrag et al., 2013). Such explanations can be explored with simple models of early diagenesis.

Most quantitative models of early diagenesis employ reactive-transport approaches that track pore-fluid chemistry and sediment composition with depth below the sediment-water interface. These models typically simulate recrystallization of primary carbonate minerals driven by diffusing or advecting seawater, with evolving water-rock ratios that alter both mineralogy and δ^{13}C (Banner and Hanson, 1990; Ahm et al., 2018). These previous approaches can explain how isotopic composition may vary along a single flow path, but do not explicitly consider the role of sedimentary transport in the accumulation of ancient sediments and the preservation of primary and diagenetic geochemical signatures.

Here we present a simple model of diagenesis and sediment transport to examine how basin-scale processes can reproduce highly negative $\delta^{13}C_{carb}$ values and sequence stratigraphic patterns that match the Shuram without changing the composition of the open marine DIC reservoir. The model includes a process similar to the precipitation of early diagenetic authigenic carbonates derived from organic carbon respiration, their reaction with existing carbonate strata, and subsequent redistribution through sediment transport and sea level driven facies migration. Our forward model provides a framework for understanding the development of extreme negative excursions and enables direct comparison with lithostratigraphic and geochemical data from global Shuram-bearing successions (Fike et al., 2006; Bergmann et al., 2025).

2. Methods

We simulate the evolution of a carbonate system along a shelf to slope transect, tracking the accumulation and respiration of organic matter within sediments, as well as diagenetic reactions between

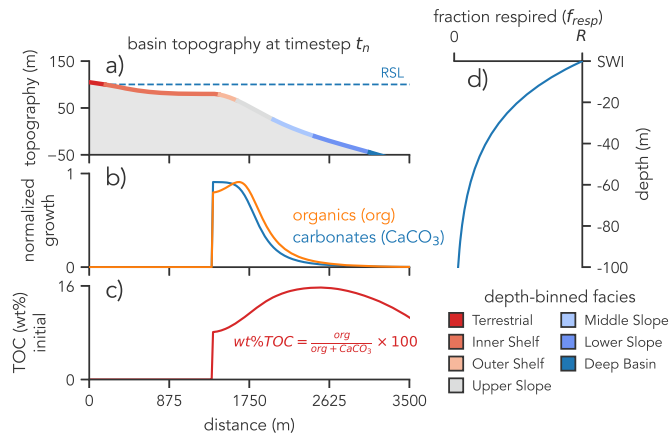

Fig. 1. Model initialization illustrating the prescribed spatial structure of sediment production and organic carbon preservation. (a) Basin topography at a single model timestep showing defined facies belts and relative sea level (RSL). (b) Prescribed depth-dependent functions for carbonate production (blue) and organic matter accumulation (orange) and how they vary with distance from shore (controlled by this time step's bathymetry). Organic matter production is defined to peak at greater water depths than carbonate growth and is scaled such that regions with lower carbonate sedimentation preserve less organic carbon. These forcings are tuned to produce the desired TOC profile (c) preserved in newly deposited sediment, calculated as $\mathrm{TOC} = \frac{\mathrm{org}}{\mathrm{org+CaCO_3}} \times 100$. TOC values are lowest on the shallow platform, increase to a maximum in slope environments, and decline toward the deep basin, consistent with global patterns of organic carbon burial (e.g., Diesing et al., 2021, Fig. S2). This spatial distribution reflects the combined influence of organic carbon supply and preservation, with slope environments receiving elevated organic carbon flux associated with nutrient delivery and upwelling, shallow shelves experiencing limited preservation, and deep basins characterized by lower productivity and sedimentation rates. (d) Prescribed fraction of organic carbon respired as a function of depth below the sediment-water interface (SWI), showing rapid remineralization near the SWI and declining respiration with burial depth. TOC preservation is modulated by both carbonate sedimentation (see Fig. S1) and rapid respiration near the sediment-water interface. Thus, shallow shelf and subaerial environments, which experience low carbonate burial efficiency (e.g., low sedimentation rates), preserve the lowest TOC values.

respiration-derived DIC and carbonate sediments. This model framework lets us examine the expected sequence stratigraphic patterns of excursions driven by reactions between respired organic carbon and carbonate sediments. We isolate this process in our simulations by holding the isotopic composition of DIC in seawater constant across both space and time.

To model accumulation, transport, and erosion in our simulated basin, we use a diffusion-based model of sediment transport that tracks the advection of conservative tracers (van Wieren et al., 2026). Carbonate sediments precipitate following light-limited production curves that simulate photosynthetic calcifiers, producing a broad shallow water shelf similar to tropical carbonate systems observed today (Fig. 1a). To track the total organic carbon (TOC) content of carbonate sediment, we simulate marine organic carbon rainout alongside carbonate sedimentation within the basin. In the model, organic carbon accumulates and is transported with carbonate sediments, but with differing preservation.

We invoke two assumptions about the deposition of organic matter in our simulated basin. First, organic carbon accumulation rates scale with carbonate (total) sedimentation rates (see Fig. S1) as modern studies demonstrate increased burial efficiency of organic carbon in zones of high sedimentation (e.g., Müller and Suess, 1979; Canfield, 1994; Arthur et al., 1998; Burdige, 2007). Specifically, organic carbon production is scaled linearly with the local carbonate sedimentation rate, which we use as a proxy for burial efficiency, such that areas of higher carbonate accumulation preserve proportionally more organic matter.

Table 1
Constants and initial conditions for the transport and early diagenesis model.

Variable	Description	Value	Reference
t	Model runtime	20 Myr	–
Δt	model timestep	0.1 kyr	–
$\epsilon_{carb-sw}$	δ^{13}C fractionation between DIC and carbonate mineral	0‰	–
$\delta^{13}C_{carb}^{primary}$	Isotopic value of primary Ediacaran sediment	0‰	Hoffman and Lamothe (2019)
ϵ_p	δ^{13}C fractionation due to photosynthesis	–25‰	Kump and Arthur (1999)
$TOC_{initial}$	initial total organic carbon content for carbonate sediments required to generate –12‰ $\delta^{13}C_{carb}$	16%	–
R	reaction rate constant	7.5%/Myr	Baker et al. (1982), Higgins et al. (2018), Ahm et al. (2018)
ρ_s	density of primary sediment	1.8 g/cm^3	Ahm et al. (2018)
V	box volume	1 m^3	–

Second, we prescribe a spatial pattern of organic carbon accumulation (Fig. 1c), with maximum accumulation in slope settings where coastal upwelling enhances primary productivity and drives a higher organic carbon flux to the seafloor. Moving basinward, accumulation declines gradually with depth as both productivity and sedimentation rates decrease (e.g., Ingall and Cappellen, 1990; Inthorn et al., 2006). Such patterns have been observed, for examples, in surface sediments along a transect from continent to basin in the North Sea (Fig. S2, Diesing et al., 2021) and in an upwelling zone off of the coast of Namibia (Inthorn et al., 2006). In contrast, shallow water platform environments are modelled with minimal organic accumulation, consistent with observations from settings such as the Great Bahama Bank (Oehlert and Swart, 2014) and parts of the Great Barrier Reef (Brunskill et al., 2002), where bank-top sediments typically contain very low TOC. We note that measured TOC values in modern surface sediments (generally <3% TOC, Oehlert and Swart, 2014; Lee et al., 2019; Diesing et al., 2021) do not represent the instantaneous flux of organic carbon to the seafloor, as much of the organic carbon deposited in the upper few centimeters is rapidly respired shortly after settling (e.g., Kristensen et al., 1995). However, as the model timestep is 0.1 kyr, newly deposited sediments already have an effective age of ~100 years. Thus, the assigned initial TOC values can be interpreted as the integrated outcome of production, deposition, and remineralization processes operating on timescales shorter than the model timestep.

In our model, the deposition of organic carbon is followed by respiration and early diagenesis. The respiration of TOC is simulated with an exponential function (Fig. 1d) with depth below the sediment-water interface (SWI) whereby a fraction of available TOC (0 to 100%) is lost to respiration with each model timestep. The fraction of organic carbon lost to early diagenesis (f_{resp}, Fig. 1d) is determined by a reaction rate term (R, %/Myr) whereby for each timestep, a percentage of the existing organic content in sediments is lost. We use $R = 7.5$%/Myr, within the range of published early diagenetic recrystallization rates (1-10%/Myr, Higgins et al. (2018) Table 1). This reaction constant is intended to capture the refractory tail of TOC loss following rapid early recycling (Arndt et al., 2013). In agreement with other diagenetic models (Boudreau and Ruddick, 1991), the fraction of TOC lost in our simulations is maximized in the upper ~50 m of the sediment column and rapidly decays with depth. With each model timestep, organic matter is both transported and respired, and TOC concentrations are updated iteratively as the basin accumulates sediment. The amount of respired TOC is estimated by subtracting the final TOC content of each grid cell from the initial TOC that accumulated in that cell. This estimate does not distinguish between organic carbon that respired in place and that which was transported away. However, sensitivity tests, where respiration is disabled indicate that the transport component of TOC is negligible, and most TOC is sourced from rainout from the water column, from when the grid cell was at the sediment-water interface. To assess the influence of respired organic carbon on the stratigraphic expression of early diagenesis, we model the decay of one mole of total organic carbon (TOC) to produce one mole of dissolved inorganic carbon (DIC) with a δ^{13}C value of –25‰ (Table 1), which is subsequently converted to authigenic carbonate cements. This formulation assumes all organic carbon that remineralizes in the sedimentary system becomes carbonate, and we report the resulting bulk carbonate isotopic composition for those sediments. While 100% conversion of organic matter to carbonates may be higher than rates observed in nature (experimental and modelling studies suggest conversion efficiencies closer to ~22%; Hu et al., 2023), only the magnitude of the alteration is sensitive to this choice, not the general stratigraphic patterns that the model produces.

Respired organic carbon is used to drive diagenetic change in the δ^{13}C of the coincident carbonate sediments. We calculate the evolving carbonate δ^{13}C composition as a mixture of organic carbon sourced DIC and DIC derived from the existing carbonate stratigraphy. For each column of stratigraphy in the simulated basin, we calculate the fraction of carbonate sediment (the fraction that is not TOC) and convert this to moles of carbonate-derived carbon. We calculate the isotopic composition of the combined organic- and carbonate-derived carbon by mass balance, representing the most negative excursion achievable in our modelled basin. This end member case assumes that organic carbon respiration is not oxidant limited, and 100% of the respired organic carbon reacts with surrounding sediment, for example through the precipitation of early diagenetic cements. Importantly, this end member approach will predict larger effects than a model that simulates a more complicated reaction framework, such as dissolution-reprecipitation or mineral-fluid exchange. Therefore, these model results show the effects when early diagenesis is maximally efficient at changing bulk δ^{13}C values. Finally, our model assumes that the δ^{13}C composition of organic matter ($\delta^{13}C_{org}$) tracks marine DIC (0‰; Table 1) and does not undergo additional modification. With $\delta^{13}C_{DIC}$ held constant, $\delta^{13}C_{org}$ remains invariant and does not change with progressive oxidation.

To simulate the sequence stratigraphic patterns observed during the Shuram excursion (Husson et al., 2015b; Busch et al., 2022; Bergmann et al., 2025), we run the model over a 20 Myr interval designed to include pre-Shuram, syn-Shuram (spanning approximately 10 Ma matching durations from Rooney et al., 2020), and post-Shuram sedimentation, and impose a sea level forcing characterized by a rapid rise followed by a gradual fall (inset, Fig. 3e). This produces a stratigraphically thin transgressive unit overlain by a thicker regressive package (Fig. 2b), allowing us to directly compare model outputs with Shuram-bearing strata using sequence stratigraphy. Stratigraphic "facies" are generated by binning carbonate sediments by depositional water depth ranges. Finally, to generate δ^{13}C values with magnitudes approaching those observed for the Shuram, we parametrize maximum TOC values of ~16%, which is the lowest value required to generate –12‰ carbonate sediments (Fig. 4a, Table 1).

3. Results

The model results in three sequence stratigraphic packages: a thin regressive sequence associated with sea level fall, a thick transgressive package associated with rapid sea level rise, and a final, thick regressive sequence associated with gradual sea level fall. Stratigraphic sections extracted from our modelled basin show an excursion ranging from approximately 0 to –12‰ (Fig. 2b; location indicated on Fig. 3d,e). During the initial regression, δ^{13}C values of sediments closely match seawater

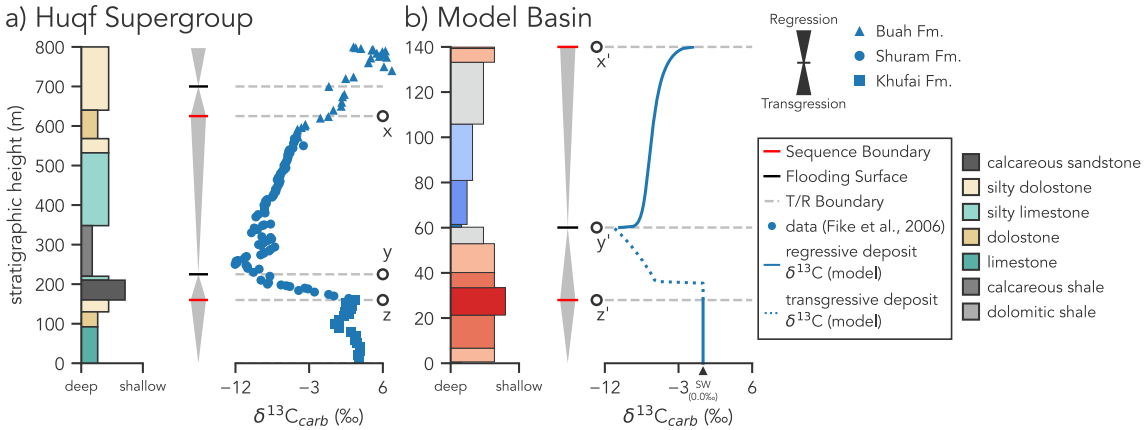

Fig. 2. Compilation of both lithostratigraphy and sequence-stratigraphy from the Huqf Supergroup in Oman (Fike et al., 2006; Busch et al., 2022; Bergmann et al., 2025) (a). Data comprise the Khufai, Shuram, and Buah Formations (in stratigraphic order). Transgressive (T) - regressive (R) sequences are shown with grey triangles, and bounded by black flooding surfaces or red sequence boundaries. Lithostratigraphy from our model simulations are shown in (b) and are color coded according to Fig. 1a. Note that the minimum values of the Shuram excursion (both in data and modelled results) occur within a transgressive sequence, followed by the gradual recovery during a regressive sequence, a pattern replicated in the simulated model basin. Suggested sequence stratigraphic correlations are shown between the Huqf data and our model basin with x,y,z and x',y',z' respectively. Lithologies shown in the legend are from Fike et al. (2006).

DIC (prescribed as 0‰). Across an erosive surface marking the shift from falling to rising sea level, $\delta^{13}C$ values in the overlying transgressive unit decrease progressively, reaching a minimum of –12‰ at the top of the transgression. As sea level begins to fall again, $\delta^{13}C$ values gradually recover toward 0‰ (Fig. 2b).

Our modelled basin captures $\delta^{13}C_{carb}$ variability spanning a range from 0‰ to –12‰ (Fig. 3d). Sediments deposited in deep water environments (e.g., lower through upper slope, Fig. 3a) have more negative $\delta^{13}C_{carb}$ values than those of shallow water environments (outer and inner shelf, Fig. 3a). Further, deep water sediments have variability in $\delta^{13}C$ comparable to that of shallow water sediments (Fig. 3a). While the minimum $\delta^{13}C$ values in individual stratigraphic columns occur at the top of the transgression, the most negative values (–12‰) appear in thin packages restricted to the distal slope setting (Fig. 3d,e).

4. Discussion

Our simulations offer a simple solution to the problem of highly negative carbon isotope excursions, values that are lower than the canonical mantle isotopic composition. By incorporating sediment transport and early diagenesis, the model reproduces first-order features of such excursions without requiring globally synchronous shifts in oceanic DIC. Instead, these patterns can emerge through broadly coincident development of similar diagenetic environments along productive continental margins. Here, we consider how this mechanism helps explain the magnitude, structure, and distribution of Ediacaran excursions such as the Shuram.

4.1. Stratigraphic trends

Isotopic profiles from our model resemble those from well studied Ediacaran successions that host the Shuram excursion, including the Huqf Supergroup of Oman, a well preserved and stratigraphically continuous Ediacaran succession (see Fig. 2a; Fike et al., 2006; Bergmann et al., 2025). Both the model and the Huqf records show near baseline values in the lowest strata, a decrease to –12‰ during rising sea level, and recovery during subsequent fall, with isotope shifts tied to stratigraphic surfaces at the transitions between falling and rising sea level (Fig. 2). Differences in baseline values (0‰ in the model vs. ~2.5‰ in Huqf; Hoffman and Lamothe, 2019) simply reflect our assumed seawater DIC composition. The excursion spans ~140 m in the model, comparable to some Shuram-bearing sections, and required a base level change of ~200 m, well within the range inferred from the rock record (50 m in the Duoshantuo Formation to >500 m in the Wonoka Formation; Busch et al., 2022). Taken together, these agreements, achieved with few adjustable parameters, demonstrate that transport and early diagenesis can reproduce sequence stratigraphic patterns consistent with global Shuram occurrences (Fike et al., 2006; Bergmann et al., 2011; Husson et al., 2015b; Busch et al., 2022).

4.2. Spatial trends

An important prediction from our simulations is that the expression of a large carbon isotope excursion (Fig. 2b) will be spatially variable within a basin. To evaluate this prediction, we compare our model results with a global compilation of Shuram $\delta^{13}C_{carb}$ data assembled by Busch et al. (2022), which includes sections from South Australia (Wonoka Formation), northwestern Canada (Gametrail and Last Chance Formations), Oman (Huqf Supergroup), Peru (San Juan Formation), and the southwestern United States (Johnnie Formation). In the compiled data compilation, samples were assigned to generalized depositional environments (inner ramp through slope) based on sedimentological and sequence stratigraphic interpretations.

Across both the modelled basin and the global dataset, deep water sediments consistently record more negative $\delta^{13}C_{carb}$ values than shallow shelf equivalents (Fig. 3a,b, supported by Wang et al., 2016). This depth trend is not unique to the Shuram; for example, during the Steptoean Carbon Isotope Excursion in the Early Paleozoic, the lightest $\delta^{13}C$ values are likewise found in the deepest depositional environments (Li et al., 2018). In our model, high organic carbon accumulation along basin margins is prescribed, but the resulting isotopic and stratigraphic patterns depend on how these settings are juxtaposed within a sedimentary succession. During intervals of slow accumulation (transgression), organic matter remains closer to the sediment-water interface and undergoes more extensive remineralization prior to burial, generating pore fluids with low $\delta^{13}C$ values of DIC. In contrast, during intervals of rapid accumulation and erosion (regression), organic matter is buried more quickly and remineralization is less effective. When these sequences are superimposed in the stratigraphic record, the outcome is a spatially variable negative $\delta^{13}C$ excursion, consistent with observations from multiple Ediacaran basins (Fig. 3).

Our simulation shows that excursion magnitude increases from land toward the basin, leading deeper environments to record more negative $\delta^{13}C$ values than shallower ones (Figs. 3a,d). This facies-dependent

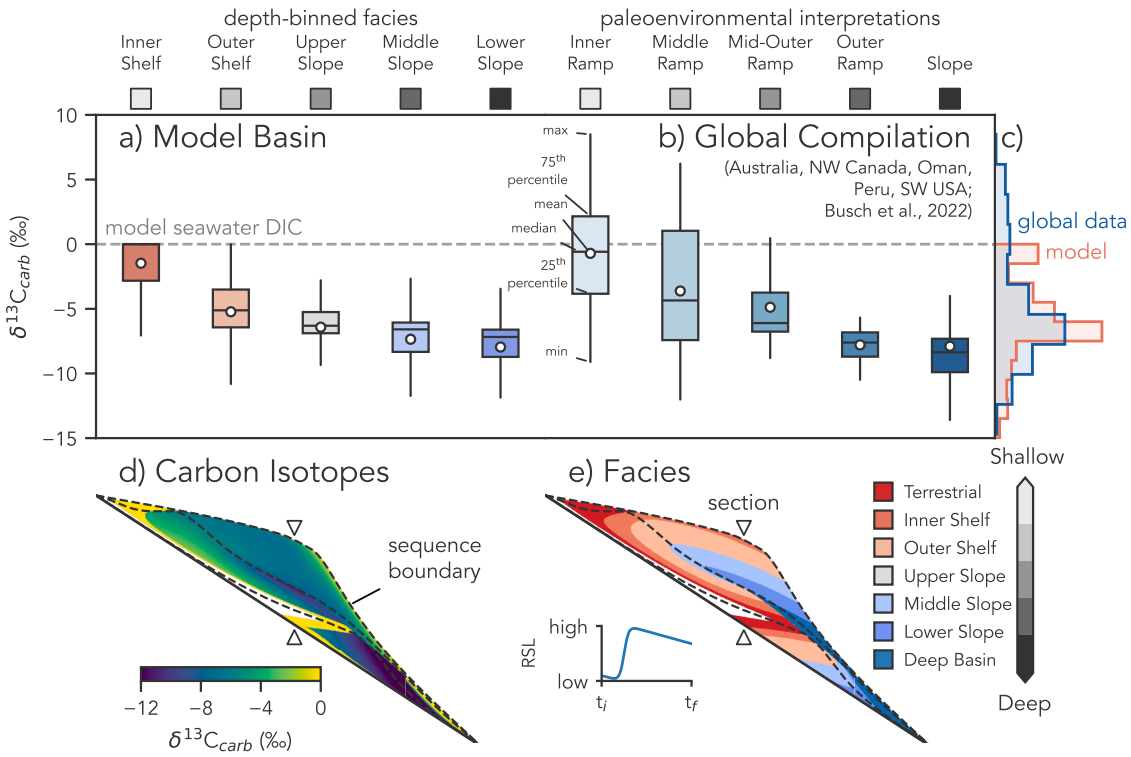

Fig. 3. Boxplots of modelled (a) and compiled (Busch et al., 2022, b) $\delta^{13}C_{carb}$ values for the Shuram excursion binned by depositional water depth, illustrating geochemical trends across a shelf-to-slope transect. Data include pre-, syn-, and post-Shuram strata and are grouped by both interpreted water depth setting and sequence stratigraphic context (i.e., transgressive vs. regressive sequences). Histograms of $\delta^{13}C_{carb}$ values for the modelled basin (restricted to the facies shown in the boxplots) and the global dataset are shown in (c). Basin-scale plots of modelled $\delta^{13}C_{carb}$ values and corresponding depositional water depth bins (facies) are shown in (d) and (e), respectively. Sequence boundaries are indicated by dashed lines and the relative sea level curve prescribed to generate the basin is shown in the inset within (e). The location of the stratigraphic section shown in Fig. 2b is indicated by white triangles. The generalized water depth (shallow to deep) for both modelled facies (a) and global facies (b) are shown with grey to black boxes for clarity as indicated by the colourbar.

gradient emerges from the stratigraphic stacking of transgressive and regressive intervals, with deeper settings preferentially recording the isotopically light end member and shallow shelves preserving values closer to primary seawater DIC. As a result, the magnitude of the excursion varies predictably with depositional setting, rather than being uniformly expressed across the basin. Middle-platform settings fall between these two end members: sedimentation is slow enough to allow remineralization, but accumulation is continuous enough to preserve a thick and coherent record. As a result, they capture the full rise and fall of excursion patterns within the basin architecture (Fig. 3d,e). In this way, the model highlights why excursions appear preferentially in some depositional settings and not others. They emerge directly from the balance between organic carbon burial rates, remineralization near the sediment-water interface, and how those processes are expressed in stratigraphy.

We note two deviations between Shuram data and our simulations. Model results show variance of populations of $\delta^{13}C$ values equal between shallow and deep environments (Fig. 3a), and approaches zero in the most landward and basinward sections where modelled sediment $\delta^{13}C$ values will derive from prescribed DIC values as prescribed by our modelled TOC distribution (Fig. 1c). In contrast, the global compilation shows the clear pattern of variance maximized in nearshore environments and minimized in deeper settings (Fig. 3b). In addition, the maximum $\delta^{13}C_{carb}$ value in the shallowest environments of the global dataset is higher than in our modelled basin, reaching ~8.5‰ compared to 0‰ (Fig. 3b,c), which is a result of our choice of the $\delta^{13}C$ value of primary carbonate sediment.

These divergences in both water depth binned $\delta^{13}C$ variance and maximum values from real data may be, in part, explained by a number of processes known to occur within shallow platform bank tops that our model does not include. Meteoric fluids (Allan and Matthews, 1982), or near shore lagoons with DIC with low $\delta^{13}C$ values owing to water column organic carbon respiration (Patterson and Walter, 1994) can contribute to variability in the $\delta^{13}C$ value of carbonate sediments. Further, processes like diurnal cycles in productivity (Geyman and Maloof, 2019), geothermal driven platform fluid-flow (Kohout, 1965), and mineralogical differences (e.g., Higgins et al., 2018) can impart more variability in $\delta^{13}C$ values of shallow carbonates. Our model does not account for such processes (the original $\delta^{13}C$ of precipitated carbonate is invariant), and as such, will only reflect values associated with the one forcing provided: the transport of TOC and its respiration during early diagenesis.

Another process not represented in our model is deep burial diagenesis. In slope to deep-basin successions, finer grain size and greater reactive surface area promote isotopic homogenization and recrystallization over tens to hundreds of meters (Pingitore, 1982), reducing primary isotopic variability in deeper water sediments. Low sedimentation rates and condensation further increases time averaging represented by stratigraphic samples and smooths short-term variability, whereas heterogeneity from meteoric alteration or localized productivity in shallow platforms may persist through burial, yielding higher isotopic variability in samples from these settings.

While global compilations demonstrate the broad basin scale patterns in $\delta^{13}C_{carb}$ expression (Fig. 3), individual basins such as the Huqf

Supergroup of Oman display little δ^{13}C variability across both a broad range of localities and depositional environments (e.g., Le Guerroué et al., 2006). Here, our model expects predictable intra-basin patterns in excursion magnitude as discussed previously, with shallower environments having dampened magnitude relative to deeper environments. We attribute such differences to processes not explicitly modelled here, mentioned above, that can reduce or obscure primary spatial gradients. These include stratigraphic correlation bias in the absence of geochronological constraints, whereby sections containing high-magnitude δ^{13}C excursions are preferentially correlated, as well as diagenetic modification by meteoric fluids. In particular, as the model generates a wide range of excursion magnitudes within a single basin, correlations based primarily on highly negative δ^{13}C values may preferentially align and emphasize stratigraphic sections from similar diagenetic facies belts, yielding composite records with less apparent intra-basin variability. As such, limited intra-basin variability in some successions does not necessarily contradict our proposed framework, but likely reflects overprinting and homogenization processes beyond the forcings explored here.

Importantly, our model isolates the first order effects of organic carbon respiration during early burial and predicts a systematic relationship between depositional environment and the mean δ^{13}C value of carbonates across the differing sequence stratigraphic contexts within a basin. The discrepancy between modelled variance patterns and that of global compilations indicates that additional shallow water processes likely contribute to the spread of δ^{13}C values within individual facies. However, enhanced variance does not negate the presence of an underlying relationship between mean $\delta^{13}C_{carb}$ values and depositional environment. Rather, such processes likely superimpose higher-order variability on a first-order trend that emerges from differences in residence time and diagenetic reaction extent.

4.3. Local drivers of the Shuram excursion

The stratigraphic and spatial trends in δ^{13}C from our modelled basin reproduce key features of the Shuram excursion observed across multiple global localities. This similarity suggests that localized processes (specifically slope-dominated productivity, coupled with remineralization of organic carbon and subsequent authigenesis) could plausibly generate δ^{13}C excursions comparable in magnitude and structure to those recorded globally. These results highlight that at least part of the Shuram signal could arise from basin-scale processes, meaning that excursions need not reflect a single, globally synchronous shift in the δ^{13}C value of a well mixed DIC reservoir, but could instead represent broadly coeval expressions of similar processes in multiple basins. We propose that widespread development of productive slope margins, and the transgressive transport of remineralized organic carbon during a global transgression, could generate a globally synchronous excursion that is decoupled from secular perturbations in the global DIC reservoir (Busch et al., 2022). Although several model parameters, such as where in the basin maximum TOC accumulation occurs and reaction rate terms, can be varied, these adjustments result in only minor changes to excursion morphology and nadir magnitude. The central outcome remains that isotopically light carbon generated in deep water environments and transported shoreward during transgression produces a negative δ^{13}C excursion, and this result is consistent across the tested parameter space (Fig. S4). Additionally, as the composition of modelled $\delta^{13}C_{org}$ is invariant while $\delta^{13}C_{carb}$ is progressively altered through diagenesis, covariation between $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$ is not expected under our model. Moreover, if $\delta^{13}C_{org}$ were diagenetically modified, differing alteration pathways and sensitivities for organic versus carbonate carbon would likely also decouple $\delta^{13}C_{carb}$ and $\delta^{13}C_{org}$, consistent with observations from the Shuram globally (e.g., Bergmann et al., 2025).

Several studies have proposed an authigenic origin for the Shuram excursion (e.g., Macdonald et al., 2013; Schrag et al., 2013) based on the global association of negative $\delta^{13}C_{carb}$ excursions with carbonate phases interbedded with dominantly siliciclastic strata. If carbonates with low δ^{13}C values precipitated as authigenic phases, such phases might be expected to be identifiable petrographically or to preserve micro-scale isotopic gradients. However, many Shuram-bearing successions preserve pristine carbonate fabrics at both macro- and micro-scales, including pristine ooids from multiple localities (e.g., Bergmann et al., 2011, 2025). The preservation of such fabrics is often interpreted as evidence for primary geochemical compositions. However, studies have demonstrated that visually pristine fabrics frequently exhibit pervasive diagenetic overprinting (e.g., Cisneros-Lazaro et al., 2022). Thus, the absence of obvious petrographic alteration does not preclude modification of bulk-rock $\delta^{13}C_{carb}$ values through diagenetic processes. Other studies argue that the lack of grain-scale isotopic gradients across ooids and other carbonate grains from China and Oman (Cui et al., 2021; Bergmann et al., 2025), coupled with carbonate phases from Australia that have high $\delta^{13}C_{carb}$ values (>0‰; Husson et al., 2020) suggest that isotopically distinct authigenic cements, where present, may record localized conditions rather than the basin-scale process responsible for the negative excursion. These observations place important constraints on the style of diagenesis responsible for the excursion, but do not rule out the influence of authigenesis.

The absence of isotopically depleted phases does not necessarily preclude a respiration-derived authigenic contribution to bulk-rock $\delta^{13}C_{carb}$ values, particularly if isotopic modification occurred after burial and was mediated by pore-fluid DIC. In this framework, isotopic signals are integrated over length scales governed by diffusion or advection rather than individual grains or beds, such that early cementation or recrystallization may incorporate isotopically light DIC without generating petrographically distinct phases or preserving micro-scale isotopic gradients (e.g., Schrag et al., 2013). Moreover, studies of Triassic carbonate successions with large negative $\delta^{13}C_{carb}$ excursions show that authigenic carbonate can occur as microcrystalline or peloidal phases that are difficult to distinguish petrographically from primary sediments (Jiang et al., 2019). Alternatively, erosion and redistribution of authigenic carbonate phases could generate carbonate grains that are subsequently incorporated into younger deposits, where they may be similarly difficult to differentiate from primary fabrics (Husson et al., 2020). Further, redistribution of this nature would likely serve to homogenize $\delta^{13}C_{carb}$ values within these sedimentary fabrics. The incorporation of such remobilized, diagenetically-derived carbonates in Ediacaran-aged sediments could contribute to the differing stratigraphic expressions observed among Shuram-bearing successions, without necessarily disrupting primary sedimentary fabrics.

4.4. Accumulating organic-rich Ediacaran sediments

The generation of highly negative δ^{13}C excursions in our model requires unusually high levels of TOC in basin sediments. As shown in Fig. 4a, the minimum initial TOC content that must accumulate in surface sediments to generate –12‰ carbonate sediments is 16% for an initial $\delta^{13}C_{DIC}$ value of 0‰ and 38% for an initial value of 5‰. These values are calculated assuming complete conversion of respired organic carbon to authigenic carbonate cements. Under lower conversion fractions that are more representative of natural systems, the TOC contents required to generate comparably negative carbonate values would be proportionally higher (see dashed lines, Fig. 4a). Though uncommon in the modern ocean, there are environments known to host similarly high TOC values. Slope sediments of the oxygen minimum zone off the Peru margin reach up to 16% TOC (Arthur et al., 1998), sediments from the Benguela upwelling system off southwest Africa reach up to 15% (Inthorn et al., 2006), and sediments from the Black Sea Romanian shelf also show values as high as 15% (Lazar et al., 2012). These regions share three characteristics that promote unusually high organic carbon contents: (1) strong upwelling that enhances productivity, (2) oxygen minimum zones that reduce aerobic respiration, and (3) high sedimentation rates that promote preservation (Burdige, 2007). Globally distributed TOC data from modern surface sediments are much lower than these

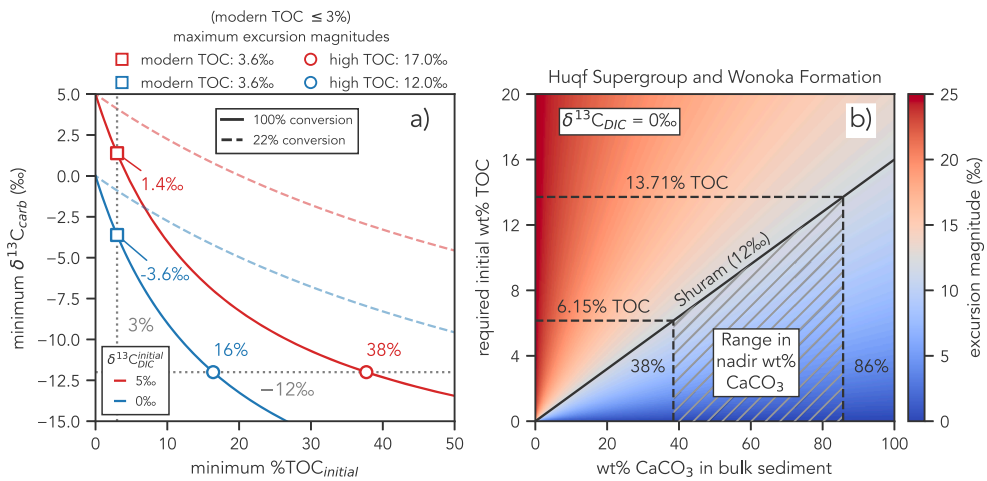

Fig. 4. (a) Plot of the minimum initial total organic carbon (%TOC) required to generate highly negative δ^{13}C values of marine carbonates given primary seawater compositions of 5‰ (red) and 0‰ (blue). Calculated values assume complete reaction between the existing carbonate and the respired organic carbon. Circles indicate %TOC required to generate –12‰ carbonate sediments, with total excursion magnitudes of 12 and 17‰ for initial DIC values of 0‰ and 5‰ respectively. Squares indicate the maximum values from modern ranges of TOC (0 to 3%; Lee et al., 2019) and the maximum excursion magnitudes ~3.6‰ that can be generated as predicted by mass balance. Dashed lines represent calculated values assuming incomplete reaction between carbonate and respired organic carbon (here 22% conversion) for reference. In this second case, under modern TOC ranges of ~3‰, excursion magnitudes are approximately 1‰. (b) Plot illustrating how the required initial TOC content (wt%) varies as a function of the wt% $CaCO_3$ in the bulk sediment participating in diagenetic reaction (the remaining material is inferred to be inert siliciclastic material) under complete conversion fractions. The solid line on panel (b) highlights combinations of TOC and carbonate weight percent values capable of producing a Shuram-magnitude excursion (12‰) with an initial $\delta^{13}C_{DIC}$ value of 0‰. Compiled ranges for wt% $CaCO_3$ values observed in carbonates containing Shuram nadir $\delta^{13}C_{carb}$ values from the Huqf Supergroup of Oman (Lee et al., 2013) and for the Wonoka Formation of Australia (Husson et al., 2015a) are shown in (b) with vertical dashed lines. These ranges correspond to initial wt% TOC values (now converted entirely to $CaCO_3$) of 6.2% to 11.9%, and 6.2% to 13.7% for the Huqf and Wonoka respectively.

ranges, with values generally between 0 to 3% (Lee et al., 2019). For example, sediments from Australian continental margins are typically below 2% TOC (Radke et al., 2017), and sediments from the North Sea are mostly less than 2.5% (Fig. S2, Diesing et al., 2021). Given these modern ranges, which represent margins without exceptional productivity or preservation conditions, the maximum decrease in $\delta^{13}C_{carb}$ from primary values achievable with our model is about 3.6‰ in pure carbonate sediments (Fig. 4a). Thus, negative excursions throughout Earth history with magnitudes ≤ 3.6‰ could be explained by such a mechanism without invoking unusually productive continental margins or changes in the availability of oxygen in seawater.

Our modelled mass-balance framework is initially formulated as an end member case in which respired organic carbon reacts with pure carbonate sediments. However, Shuram-bearing successions have been observed to contain a high proportion of siliciclastic material. Both the Huqf Supergroup of Oman and the Wonoka Formation of South Australia include significant siliciclastic fractions within their lithologies rather than pure carbonates (Fike et al., 2006; Husson et al., 2015b). Incorporating mixed carbonate-siliciclastic sediments in our model alters the effective size of the carbonate carbon reservoir participating in diagenetic reaction. As illustrated in Fig. 4b, reducing the bulk wt% $CaCO_3$ in the reacting sediment (with the remaining fraction treated as inert siliciclastic material) lowers the amount of organic carbon required to generate a Shuram magnitude excursion of 12‰ (solid line, Fig. 4b). For example, measured values of wt% $CaCO_3$ for carbonates containing nadir Shuram δ^{13}C values from the Huqf Supergroup (Lee et al., 2013) range from 38% to 75% which, by our mass balance, requires initial TOC values of ~6-12% (Fig. 4b). Similarly, ranges from the Wonoka Formation of South Australia (Husson et al., 2015a) yield TOC values of ~6-14% (Fig. 4b). Thus, the 16 wt% TOC (now converted entirely to $CaCO_3$) requirement derived for a now pure carbonate sediment (Fig. 4a) represents a conservative upper bound. Mixed sediments, characteristic of many Shuram-bearing successions, can achieve Shuram-scale excursions with substantially lower initial TOC contents. In fact, calculations of primary Ediacaran TOC content (pre burial loss) during the Shuram range from 0.1 to 16.4% with mean values of 3.3% over the Shuram bearing members of the Doushantuo Formation, attributed to enhanced productivity driven by coastal upwelling (Cañadas et al., 2024), consistent with our model.

Beyond sediment composition, an additional upper-bound constraint in our model framework arises from interaction with the overlying seawater reservoir. Surface sediments undergoing anaerobic respiration at or near the sediment-water interface not only undergo the most extensive diagenetic reaction, but are also subject to the greatest influence of seawater buffering (Higgins et al., 2018). Advective exchange of seawater that is isotopically enriched relative to diagenetic pore fluid DIC acts as a buffer, reducing the magnitude of organic influence on carbonate δ^{13}C values. In the context of our model results, such buffering would be expected to dampen excursion magnitude rather than alter the direction or sequence stratigraphic expression of the signal. Moreover, as with diagenetic reaction extent, the efficiency of seawater buffering is strongly dependent on sediment accumulation rate and residence time at the sediment-water interface. More rapid burial limits advective exchange with seawater and therefore preserves a larger diagenetic isotopic overprint. Our model therefore represents an end-member case in which seawater buffering is minimized and diagenetic modification of the carbonate reservoir is maximized. Natural systems that experience sustained exchange with overlying seawater would likely yield smaller-magnitude excursions than those predicted here, emphasizing that our calculated excursion magnitudes should be viewed as conservative upper estimates of this diagenetic effect.

Taken together, these upper-bound scenarios still require TOC values that, while achievable in certain modern environments, are rare. The persistence and global expression of a Shuram-scale excursion likely reflects unique conditions of the Ediacaran ocean-atmosphere system. One hypothesis is that lower atmospheric oxygen and widespread marine anoxia in the Proterozoic promoted authigenesis (Schrag et al., 2013). Under oxic conditions, aerobic respiration consumes oxygen and produces CO_2, lowering carbonate saturation in pore fluids and suppressing the preservation of authigenic cements. In contrast, suboxic to anoxic

respiration pathways (such as sulfate reduction) increase alkalinity, raising carbonate saturation and favoring the precipitation of isotopically light authigenic carbonates (Higgins et al., 2009).

Oxygen levels during the Ediacaran are thought to be on the order of ~10-15% of present atmospheric levels (Lyons et al., 2014; Stockey et al., 2024). Constraints from marine redox proxies indicate that dissolved oxygen concentrations in marine shelf environments ranged from suboxic to hypoxic during this interval (Stockey et al., 2024). Enrichments in redox-sensitive trace metals including uranium and molybdenum over the Ediacaran-Cambrian transition (Lyons et al., 2014; Stockey et al., 2024) are commonly interpreted as reflecting progressive oxygenation of the atmosphere, and expansion of oxic conditions in the upper water column. Such shifts in surface oxygenation would not necessarily require fully oxygenated bottom waters. Instead, conditions would have favored enhanced preservation of organic matter in oxygen-limited seafloor environments, consistent with an inferred increase in TOC in the terminal Ediacaran and continuing into the Paleozoic (Stockey et al., 2024). Furthermore, limited oxygen penetration into Ediacaran sediments would have restricted aerobic remineralization pathways, promoting anaerobic respiration and elevated pore fluid alkalinity, conditions conducive to higher carbonate saturation states and the formation and preservation of authigenic carbonates in marine surface sediments (Higgins et al., 2009). Together, these factors suggest that the unusual redox state and sediment composition of the Ediacaran created the conditions necessary for generating Shuram-scale negative $\delta^{13}C_{carb}$ excursions. This line of reasoning raises the possibility that the absence of similar negative excursions in Phanerozoic strata may partly reflect broader changes in global oxygenation state, with enhanced aerobic remineralization under more persistently oxygenated conditions acting to reduce the potential for forming volumetrically important authigenic carbonates.

4.5. An increased oxidant supply to the Ediacaran ocean

The development of large magnitude negative excursions in our model requires not only unusually high TOC accumulation, but also sufficient oxidants to remineralize that organic carbon. As discussed above, aerobic respiration lowers carbonate saturation in pore fluids and inhibits authigenic carbonate preservation. Under the low-oxygen conditions hypothesized for the terminal Ediacaran, respiration of organic matter would therefore have proceeded predominantly via anaerobic pathways, most notably sulfate reduction (e.g., Shi et al., 2023). Independent geochemical evidence suggests that sulfate availability may have increased during the onset of the Shuram excursion, consistent with an increase in continental weathering inferred from rising seawater $^{87}Sr/^{86}Sr$ ratios across Shuram-bearing successions (Li et al., 2017; Sun et al., 2025). Thus, increased weathering would have supplied additional sulfate to the ocean, increasing the oxidizing capacity available for anaerobic respiration (such as bacterial sulfate reduction), while simultaneously delivering terrestrially derived nutrients to marine systems (Cui et al., 2022). Together, these inputs could promote both organic matter remineralization and elevated primary productivity within a redox-stratified ocean (Ostrander et al., 2023; Cañadas et al., 2024).

While increased oxidant supply to the Ediacaran oceans would enhance rates of sulfate reduction, our model requires the prior accumulation of a large sedimentary organic carbon reservoir to generate Shuram-magnitude (–12‰) excursions. In this framework, Shuram-scale excursions emerge only when organic carbon is both efficiently preserved and later remineralized, requiring a convergence of locally high sedimentation rates, low-oxygen bottom-water conditions, adequate oxidant availability, and sustained primary productivity. If surface waters remained sufficiently oxygenated to avoid persistent anoxia, and sustain photosynthetic production, coastal upwelling of nutrient-rich waters sourced from anoxic bottom waters could further amplify productivity and associated organic matter accumulation along slope environments. Such upwelling has been invoked to explain the occurrence of phosphorites and barite in Shuram-aged successions and is consistent with the conditions required by our model (Williams and Schmidt, 2018; Cheng et al., 2022).

4.6. Comparison to Ca & S proxy records

Independent geochemical proxy systems also place constraints on the style of diagenesis responsible for the Shuram excursion. Calcium isotope data, for example, have been interpreted to indicate that shifts in $\delta^{44/40}Ca$ during the downturn of the Shuram excursion reflect changes in dominant carbonate mineralogy and, more broadly, diagenesis occurring in a sediment-buffered environment (Husson et al., 2015a). In this framework, the isotopic composition of diagenetic carbonate is controlled largely by exchange with the primary sediment, rather than by an open-system fluid flux. Our model is consistent with this interpretation. If authigenic carbonate precipitating within pore spaces is initially aragonitic, consistent with primary Shuram sediment mineralogy inferred from Ca isotopes (Husson et al., 2015a), and if a sufficient fraction of this diagenetic carbonate is incorporated into the preserved bulk sediment, then this mechanism may offer a partial explanation for the observed trends in $\delta^{44/40}Ca$. When considering the effects of authigenesis on the sulfur isotope record, enhanced bacterial sulfate reduction would generate isotopically depleted sulfides, thus yielding authigenic phases incorporating enriched $\delta^{34}S$ values. While workers have shown the opposite pattern in some Shuram-bearing sections (e.g., Fike et al., 2006), one possible explanation within our framework is that open-system advection of seawater sulfate into pore fluids, coupled with elevated Ediacaran seawater sulfate concentrations, buffered the upper sediment column with respect to $\delta^{34}S$. Under such conditions, carbonates could plausibly precipitate with both isotopically depleted $\delta^{34}S$ and $\delta^{13}C$ compositions. However, as our model does not include sulfur cycling, this interpretation cannot be directly evaluated.

4.7. Basin-scale variance

Shuram-bearing successions show substantial variability in the magnitude and stratigraphic expression of carbonate $\delta^{13}C$ excursions, ranging from abrupt, stratigraphically condensed downturns to more gradual excursions developed over thick successions. For example, expressions of Shuram downturns in the Huqf Supergroup of Oman (Fike et al., 2006) are typically sharp and condensed, whereas sections from the Doushantuo Formation of South China (e.g., Li et al., 2017) commonly record more gradual downturns in $\delta^{13}C_{carb}$ possibly reflecting a shift in sedimentation rates. In our model, this variability arises primarily from differences in sediment accumulation rates and the time that organic-rich sediments spend near the sediment-water interface. Rapid accumulation compresses excursions over thin stratigraphic intervals, producing abrupt downturns in isotopic values, whereas slower accumulation stretches the same excursion over thicker successions and yields more gradual changes (Fig. S3). Sections with muted variability may reflect laterally restricted sampling within relatively uniform depositional environments. Where sections are measured predominantly along strike in similar facies belts, variability in Shuram magnitude would be expected to be limited. Moreover, depositional environments are commonly inferred from lithofacies that can encompass a range of sedimentological expressions despite representing broadly similar environmental conditions.

An additional issue is the circularity of correlating sections primarily on the basis of carbonate $\delta^{13}C$ records and then interpreting the resulting excursions as globally synchronous. This approach risks excluding coeval sections that lack strongly negative excursions by assigning them to intervals before or after the Shuram. The apparent global extent of the Shuram may therefore partly reflect a bias toward correlating productive slope margins together, where accumulation rates and diagenetic conditions favor strong excursion development, while sections

characterized by more typical accumulation patterns are underrepresented in global carbon isotope correlation curves.

5. Conclusions

Negative δ^{13}C excursions in marine carbonate sediments have recurred throughout Earth history, and their origin is often debated as either global perturbations to the carbon cycle or local diagenetic overprints. Our results show that variations in sediment accumulation driven by changes in relative sea level can generate early diagenetic excursions of the same magnitude and structure typically attributed to global signals. In our model, isotopically light carbonate derives from organic matter that accumulates preferentially along basin slopes and is remineralized more completely when burial is slow, producing authigenic carbonate with low δ^{13}C values that is redistributed across shallow platforms. Superposition of these different diagenetic settings through stratigraphic succession yields excursions in which values are consistently more negative during transgression, with the most negative values preserved at the top of the transgressive package. By explicitly incorporating spatial heterogeneity and facies migration, the model makes specific predictions about the relationship between relative sea level, facies and δ^{13}C values. As a case study, we demonstrate that the Ediacaran Shuram excursion can be explained without invoking changes to the global DIC reservoir. Modelled stratigraphy captures the baseline, downturn, and recovery matching sequence stratigraphic patterns observed in Shuram-bearing successions globally (Fike et al., 2006; Busch et al., 2022), with deep water facies consistently more negative than shallow shelf facies. More broadly, this framework highlights how local processes operating simultaneously across multiple basins may offer alternative explanations for extreme negative excursions over Earth history.

CRediT authorship contribution statement

Connor S. van Wieren: Conceptualization, Investigation, Methodology, Formal Analysis, Visualization, Writing – original draft; **Blake Dyer:** Writing – review & editing, Supervision, Resources, Conceptualization; **Jon M. Husson:** Writing – review & editing, Supervision, Resources, Conceptualization.

Code and data availability

All code and data required to reproduce the results of these analyses are accessible on Github (https://github.com/VanWieren/shuram_TOC_public).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.epsl.2026.120058.

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